SECTION 4
AADK TASK 4

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Course : Android App Development with Gen AI

Problem 1 Summary:

The exercise required building a UI layout where a profile image is displayed above the user's name and role using Jetpack Compose composables. The layout must be vertically centered and properly spaced. Planned Approach: • Use Column() for vertical arrangement.

• Use Image() for profile picture.
• Use Text() for name and role.
• Apply Modifier.padding(16.dp) for spacing.
• Use horizontalAlignment = Alignment.CenterHorizontally.

Reflection: The implementation helped in understanding vertical arrangement using Column. Proper spacing required use of Spacer() and padding modifiers. Alignment must be carefully handled for professional UI design.

Problem 2 Summary:

The exercise required creating a horizontal layout using Row composable with an icon and text aligned side-by-side. Planned Approach:

• Use Row() composable.
• Add Icon() and Text().
• Use Modifier.padding() and Spacer(width = 8.dp).

Reflection:

Row composable arranges elements horizontally. Spacing and alignment are important for readable UI. Planning reduces layout restructuring later.

Planning Checklist Template:

1. Understand UI requirement.
2. Sketch layout structure.
3. Choose composables (Row, Column, Box).
4. Apply modifiers (padding, size, alignment).
5. Add resources (images/icons).
6. Test using @Preview.
7. Run on multiple screen sizes.

Final Reflection:

Planning and reflection improve UI development efficiency. It ensures better layout organization and reduces debugging time. Developers become more confident in solving new Compose UI problems.

Task 1
Composables Practice Reflection & Planning

Description

This task focuses on **understanding Jetpack Compose composables** by reviewing problems from the **"Practice: Compose Basics" codelab**.
The goal is to:

- Understand how composables work
- Learn how to **plan UI before coding**
- Reflect on what you learned from solving Compose UI problems.

Practical Action Items

1. Review Two Problems from Compose Basics

You must select **at least two UI exercises** from the Compose Basics practice.
Examples:

- UI using **Column**
- UI using **Row**
- UI using **Box**

2. For Each Problem, Write Three Things

a) Problem Summary (2–3 lines)
Explain what the UI problem asked you to build.
Example:
The exercise required creating a layout where a profile image is displayed above the user's name and role using composables. The layout should be vertically aligned and centered on the screen.

b) Planned Approach (Before Coding)
Explain **how you planned the UI** before writing code.
Example:

- Use Column() to arrange items vertically
- Use Image() for the profile picture
- Use Text() composables for name and role
- Apply Modifier.padding() for spacing

c) Result / Reflection
Explain what happened **after implementing the solution**.
Example:

- Learned how Column arranges elements vertically
- Spacing required Spacer() or padding
- Alignment needed horizontalAlignment

3. Create a Planning Checklist Template

Create a **general checklist for solving future UI problems**.
Example Checklist:

1. Understand UI requirement
2. Sketch the layout
3. Select composables (Row, Column, Box)
4. Apply modifiers (padding, size, alignment)
5. Add images/icons/text
6. Test UI using @Preview
7. Run on device

4. Short Reflection (3–4 lines)

Explain **how planning helps in Compose development**.
Example:
Planning UI before coding helps developers visualize the layout structure and choose appropriate composables. It reduces coding errors and makes the development process faster and more organized.

Business Card Concept:

Student Developer Card Title: Pavan Kumar Role: Android Developer Bio: Passionate about building Android apps using Kotlin and Jetpack Compose. Contact Information: • Phone: +91 9876543210 • Email: pavan@email.com

Layout Structure:

Column { Image(Profile) Text(Name) Text(Role) Row(Phone Icon + Number) Row(Email Icon + Email) } Spacing Plan: • 16dp outer padding. • 8dp spacing between icon and text. • Center alignment for all elements.

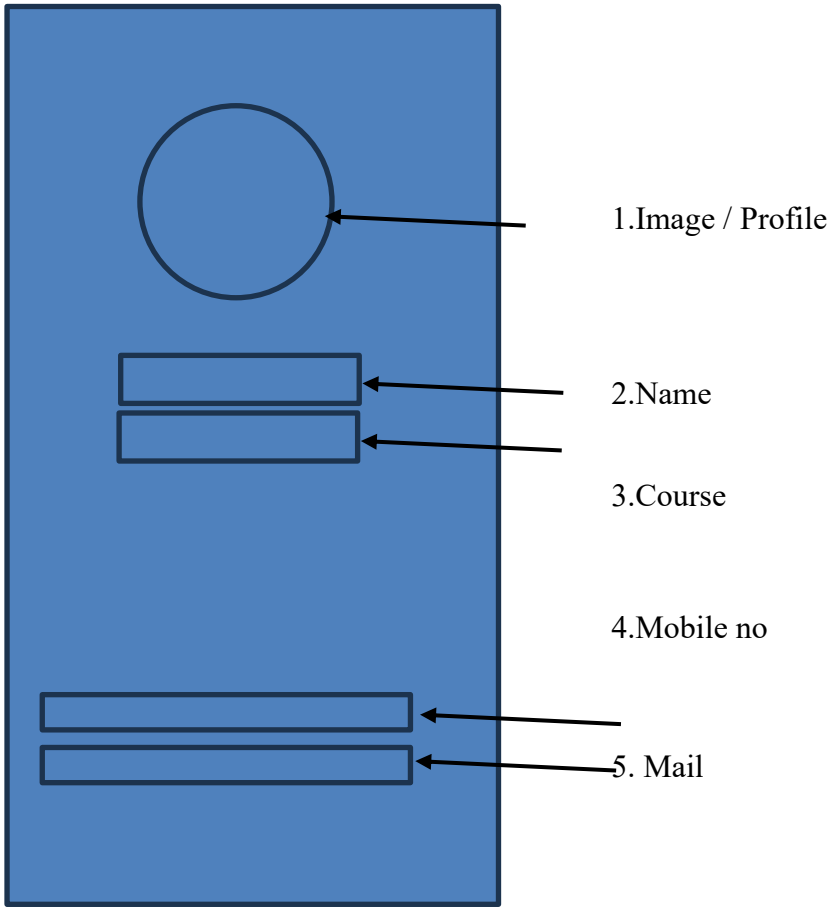
Tablet Variant Layout:

Row { Image(Profile) Column { Name Role Contacts } } This layout ensures responsiveness across devices.

Resource & Style Plan:

Image: profile.png (drawable-nodpi) Title Font Size: 28sp Subtitle Font Size: 18sp Contact Font Size: 14sp Primary Color: Blue Background Color: White Text Color: Black Accessibility Plan: • Add contentDescription to Image. • Ensure minimum touch size 48dp. • Maintain high color contrast.

Reflection: The Column composable is ideal for vertical layouts. Rows are useful for contact details. Planning UI structure before coding ensures clarity and maintainability.



Task 2
Design & Document a Business Card Layout (Pre-Code)

Description

This task focuses on **UI design planning before writing code**.
You must design your **own Business Card layout UI** inspired by the **Jetpack Compose Business Card Codelab**.

The goal is to understand:
- UI layout structure
- Composable hierarchy
- Resource planning
- Accessibility

Practical Action Items

1. Choose a Business Card Concept

Example ideas:
- **My Developer Card**
- **Student Portfolio Card**
- **Artist Promotion Card**

Example:
Title: Pavan Kumar
Role: Android Developer

2. Define the Content

Include:
Title & Subtitle
Example:
Name: Pavan Kumar
Role: Android Developer

Short Bio
Example:
Passionate Android developer specializing in Kotlin and Jetpack Compose.

Contact Information
At least **two fields**:
Example:
- Phone Number
- Email Address

Image
Example:

- Profile picture
- Logo

3. Create a Wireframe / Sketch

You must draw a **layout structure**.
Example Layout:

Column

Image (Profile)

Text (Name)

Text (Role)

Row (Phone Icon + Number)

Row (Email Icon + Email)

4. Label Composables

Exa

```
Column {
   Image()
   Text()
   Text()
   Row {
      Icon()
      Text()
   }
}
```

5. Add Modifiers & Spacing

Example:

- 16dp margin around the card
- 8dp space between icon and text
- Center alignment

6. Create a Second Layout Variant

You must also design how the UI will look for:

- **Tablet**
- **Landscape mode**

Example:
Phone Layout:
Column
Image
Name
Role
Contacts
Tablet Layout:
Row
Image | Column(Name, Role, Contacts)

7. Resource & Style Plan

Image Resource
Example:
profile.png → drawable-nodpi

Text Style

Element	Font Size
Title	28sp
Subtitle	18sp
Contact	14sp

Color Palette
Example:
Primary: Blue
Background: White
Text: Black

Accessibility Plan
Example:

- Add contentDescription to images
- Icons should have **minimum touch size 48dp**

8. Reflection

Example:
The chosen layout ensures clear visual hierarchy and readability. The Column composable allows easy vertical alignment of UI components, while Rows are used for displaying contact details.

TASK 3 – Layout Review & Improvement Plan

Evaluation Criteria: Visual Alignment: Check margins, spacing, and center alignment.
Readability: Ensure proper font hierarchy (Title > Subtitle > Contact). Maintain good color contrast.
Responsiveness: Test layout on phone, tablet, and landscape orientation. Accessibility: Add contentDescription for images. Maintain minimum touch size of 48dp.

Issue Identified	Proposed Fix	Priority
Subtitle text small	Increase to 16sp	High
Low padding	Add 16dp margin	Medium
Image too large	Compress drawable	Medium
Missing accessibility labels	Add contentDescription	High
Tablet layout not optimized	Create Row variant	Low

Final Reflection: UI evaluation improves application quality. It helps identify usability, readability, and accessibility issues. Continuous improvement ensures professional-level Compose applications.

Description
This task requires **evaluating the business card UI** and identifying improvements.
The goal is to:

- Analyze UI quality
- Improve layout
- Ensure responsiveness
- Improve accessibility.

Practical Action Items

1. Capture Screenshots

You must include **two screenshots**:

1.Phone screen layout
2.Tablet or landscape layout

2. Perform Layout Review

Evaluate using the following criteria.

Visual Alignment & Spacing
Check:

- Are elements centered?
- Are margins consistent?

Example:
Good:

- Equal padding around elements

Bad:

- Text too close to edges

Readability
Check:

- Font sizes
- Color contrast

Example:
Title should be larger than subtitle.

Resource Usage
Check:

- Image file size
- Proper drawable folders

Example:
Bad:

- 5MB image file

Good:

- Optimized PNG image

Responsiveness
Check how the UI behaves on:

- Phones
- Tablets
- Landscape mode

Accessibility
Check:

- Images have contentDescription
- Icons are clickable
- Touch targets are at least **48dp**

3. Improvement Plan

Create a table like this:

Issue	Proposed Fix	Priority
Subtitle text small	Increase size to 16sp	High
Image too large	Compress image	Medium
Low padding	Add 16dp margin	Medium
Icons missing labels	Add contentDescription	High

4. Final Reflection

Example:
Reviewing UI layouts helps identify usability and accessibility problems that may not be obvious during development. This evaluation process improves the quality of Compose applications.

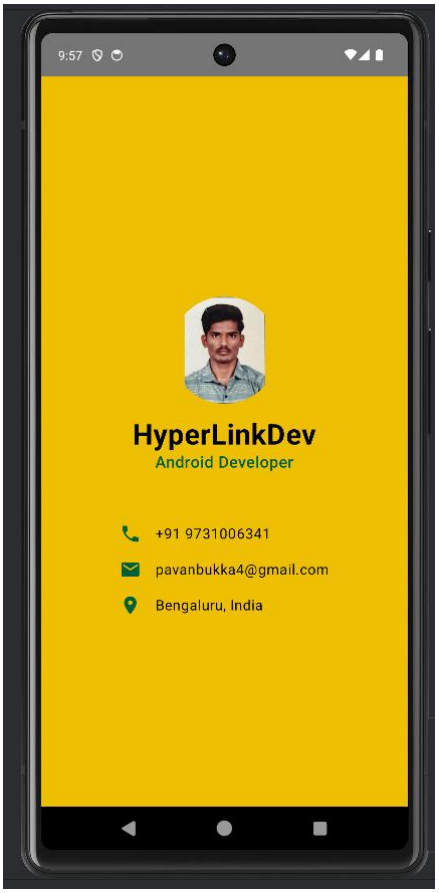
Code :

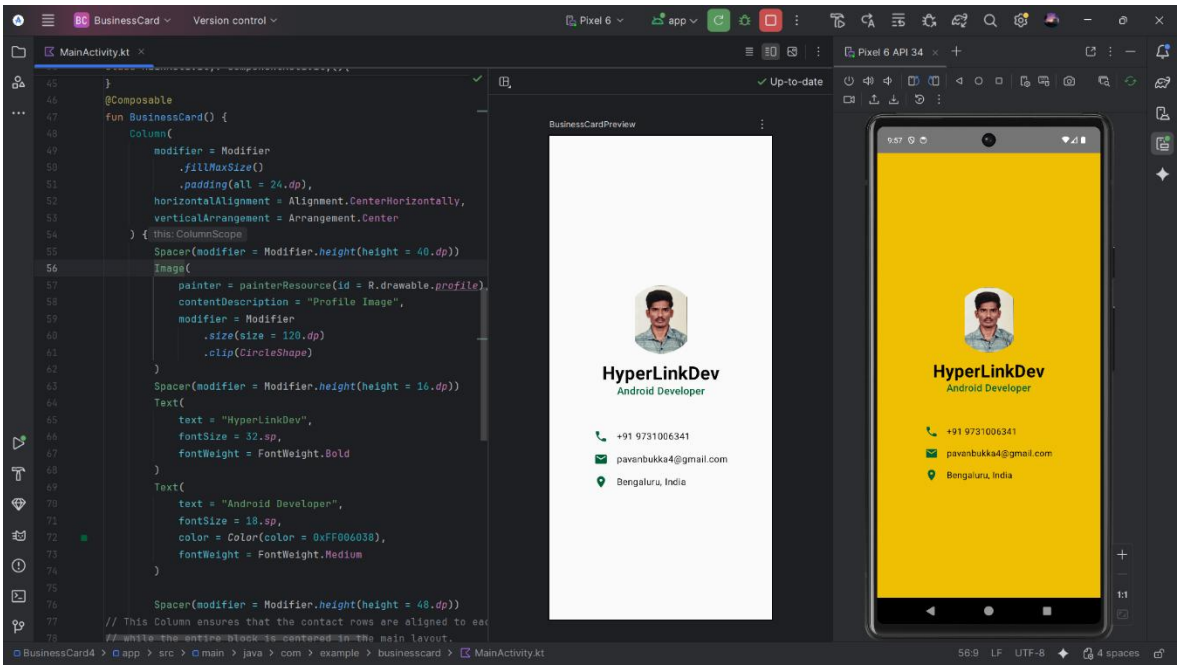
```
package com.example.businesscard

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.Image
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.shape.CircleShape
import androidx.compose.material.icons.Icons
import androidx.compose.material.icons.filled.Email
import androidx.compose.material.icons.filled.Phone
import androidx.compose.material.icons.filled.Place
import androidx.compose.material3.Icon
import androidx.compose.material3.Surface
import androidx.compose.material3.Text
import androidx.compose.runtime.Composable
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.draw.clip
import androidx.compose.ui.graphics.Color
import androidx.compose.ui.graphics.vector.ImageVector
import androidx.compose.ui.res.painterResource
import androidx.compose.ui.text.font.FontWeight
import androidx.compose.ui.unit.dp
import androidx.compose.ui.unit.sp
import com.example.businesscard.ui.theme.BusinessCardTheme
import androidx.compose.ui.tooling.preview.Preview



class MainActivity: ComponentActivity(){
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContent {
            BusinessCardTheme {
                Surface(
                    modifier = Modifier.fillMaxSize(),
                    color=Color (color =0xFFEFBF04)
                ) {
                    BusinessCard()
                }
            }
        }
    }
}
```

```
@Composable
fun BusinessCard() {
    Column(
        modifier = Modifier
            .fillMaxSize()
            .padding(all = 24.dp),
        horizontalAlignment = Alignment.CenterHorizontally,
        verticalArrangement = Arrangement.Center
    ) {
        Spacer(modifier = Modifier.height(height = 40.dp))
        Image(
            painter = painterResource(id = R.drawable.profile),
            contentDescription = "Profile Image",
            modifier = Modifier
                .size(size = 120.dp)
                .clip(CircleShape)
        )
        Spacer(modifier = Modifier.height(height = 16.dp))
        Text(
            text = "HyperLinkDev",
            fontSize = 32.sp,
            fontWeight = FontWeight.Bold
        )
        Text(
            text = "Android Developer",
            fontSize = 18.sp,
            color = Color(color = 0xFF006038),
            fontWeight = FontWeight.Medium
        )

        Spacer(modifier = Modifier.height(height = 48.dp))
// This Column ensures that the contact rows are aligned to each other
// while the entire block is centered in the main layout.
        Column(
            horizontalAlignment = Alignment.Start,
            modifier = Modifier.wrapContentWidth()
        ) {
            ContactRow(icon = Icons.Default.Phone, text = "+91 9731006341")
            ContactRow(icon = Icons.Default.Email, text = "pavanbukka4@gmail.com")
            ContactRow(icon = Icons.Default.Place, text = "Bengaluru, India")
        }
    }
}

@Composable
fun ContactRow(icon: ImageVector, text: String) {
    Row(
        verticalAlignment = Alignment.CenterVertically,
        modifier = Modifier.padding(vertical = 8.dp)
    ) {
        Icon(
            imageVector = icon,
            contentDescription = null,
            modifier = Modifier.size(size = 24.dp),
            tint = Color(color = 0xFF006038)
        )
```

```
  Spacer(modifier = Modifier.width(width = 16.dp))
        Text(
            text = text,
            fontSize = 16.sp,
            color = Color.Black
        )
    }
}
@Preview(showBackground = true)
@Composable
fun BusinessCardPreview() {
    BusinessCardTheme {
        BusinessCard()
    }
}
```

Ouyput / final result :




BusinessCard
Version control
Pixel 6
app
MainActivity.kt
Pixel 6 API 34
Up-to-date
BusinessCardPreview
}
@Composable
fun BusinessCard() {
Column(
modifier = Modifier
.fillMaxSize()
.padding(all = 24.dp),
horizontalAlignment = Alignment.CenterHorizontally,
verticalArrangement = Arrangement.Center
) { this: ColumnScope
Spacer(modifier = Modifier.height(height = 40.dp))
Image(
painter = painterResource(id = R.drawable.profile),
contentDescription = "Profile Image",
modifier = Modifier
.size(size = 120.dp)
.clip(CircleShape)
)
Spacer(modifier = Modifier.height(height = 16.dp))
Text(
text = "HyperLinkDev",
fontSize = 32.sp,
fontWeight = FontWeight.Bold
)
Text(
text = "Android Developer",
fontSize = 18.sp,
color = Color(color = 0xFF006038),
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)
Spacer(modifier = Modifier.height(height = 48.dp))
// This Column ensures that the contact rows are aligned to eac
// while the entire block is centered in the main layout.
HyperLinkDev
Android Developer
+91 9731006341
pavanbukka4@gmail.com
Bengaluru, India
BusinessCard4 > app > src > main > java > com > example > businesscard > MainActivity.kt
56:9
LF
UTF-8
4 spaces